

Satyajith Chilappagari

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WORK EXPERIENCE

AMAZON.COM, INC.

CUPERTINO, CA, USA

Machine Learning Engineer II at Annapurna Labs

July 2024 - Present

- **Extended vLLM support for Trainium and Inferentia**, implementing capabilities for standard inference, multi-modal inference, speculative decoding, quantization, and multi-LoRA using AWS Neuron.
- **Created an internal pipeline for vLLM** supporting large models (e.g., LLaMA 3.1 405B), enabling validation of open-source contributions and Neuron's feature implementations, directly supporting GenAI developer workflows.
- **Developed an automated benchmarking system** integrating vLLM and LLMPerf to evaluate cutting-edge LLM models on Trainium (Trn) and Inferentia (Inf) hardware, reducing developer time spent on benchmarks by over **90%**.
- **Built automated benchmarking pipelines** to catch Neuron inference regressions, enabling CI/CD-gated weekly benchmark reviews and cutting validation cycles from 6 weeks to 1.
- **Enabled multi-LoRA support** on AWS Neuron, empowering customers to deploy custom inference adapters on Trn1, Inf2.
- **Designed and implemented an upstream workflow** to contribute AWS Neuron vLLM enhancements from internal packages to the open-source vLLM repository, cutting release timelines from **60+ days to 2 days** and accelerating customer access to new Neuron features.
- **Enabled FP-8 support in PyTorch XLA**, allowing AWS Neuron to deliver LLM results with quantized models, reducing memory usage by 50%.

Software Development Engineer at Alexa

Feb 2022 - July 2024

- **Developed a real-time ingestion platform for Alexa Proactive Experiences** (30K+ TPS) supporting validation, transformation, and dispatch. Downstream datasets powered product analytics via Athena.
- **Designed and integrated a version-supported schema validation module**, ensuring uniform event structure across all Alexa proactive interactions.
- **Implemented pipeline-based staged rollout in production environments**, reducing impact of faulty deployments to just 1% of traffic. This reduced the blast radius of an erroneous database change that would have caused a 3-day data loss.
- **Reduced costs by 40% per month** by replacing AWS Firehose with a custom module that batches and writes data in Parquet format to S3.
- **Managed pipeline dependencies with Apache Airflow**, improving diagnosis of failures in multi-stage dataset flows. DAG-based visibility accelerated triage and resolution for internal analytics teams.

DRESSLIFE GMBH

HANNOVER, GERMANY

Backend Developer

Mar 2017 - Dec 2019

- **Engineered the back-end of a fashion-based mobile application** using an event-driven architecture based on Apache Kafka, Cassandra and Elasticsearch.
- **Developed an innovative size recommendation module** leveraging customers' height and weight data to simulate bodies.
- **Built a personalized style recommendation engine** using autoencoders and convolutional neural networks (CNNs) to reduce body and clothing features into embeddings and perform nearest-neighbor retrieval in the learned space.
- **Deployed the system in production for two German clothing brands**, resulting in a 5% revenue increase by improving fit accuracy and boosting conversion through more personalized suggestions.

EDUCATION

RUTGERS, THE STATE UNIVERSITY OF NEW JERSEY

NEW BRUNSWICK, NJ, USA

MS in Computer Science (Specialization: Machine Learning) – GPA 3.90/4

- **Part-time lecturer and grader:** Taught a university level Data 101 course using R as the primary language.
- **Internship at Commvault (June 2021 - December 2021):** Secured second place among 70 interns in the intern project competition for developing an NLP-based bug-search tool.

MANIPAL UNIVERSITY

MANIPAL, INDIA

B Tech. in Computer Science and Engineering

July 2013 - July 2017

SKILLS

Programming languages: Python, Java, TypeScript, C++, C, JavaScript.

Databases: DynamoDB, Cassandra, SQL, Elasticsearch, MySQL, MongoDB.

Frameworks, Cloud services and Visualization tools: PyTorch, vLLM, Apache Kafka, Docker, Flask, Airflow, Git, Grafana.

AWS - EC2, ECS, Lambda, S3, RDS, DynamoDB, CloudFormation, CloudWatch, Kinesis, Athena.